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Von Mises Versus the Bookies; An Analysis of Hypothetical Frequentism

In his 1957 publication *Probability, Statistics, and Truth*, Richard von Mises argues for a hypothetical frequentist interpretation of objective probability. Von Mises' line of reasoning concludes that the objective probability for events is arrived upon by properly defining a collective and subsequently finding the limiting frequency of the event as extrapolated from an analysis of a hypothetically infinite collective. Although von Mises' interpretation finds success in alleviating some of the glaring issues of actual frequentism, the strict requirements of a definition of probability advocated for by von Mises' and his stubborn adherence to such dogma disallows the possibility of implementing probabilistic analysis in singular cases, calling into question the adequacy of his interpretation. I seek to reconstruct von Mises' argument through the premises that support it and further investigate the implications of a frequentist account which cannot lend itself to single-case probabilities. First, I will iterate over von Mises' definitions for collective, attribute, unlimited repetition, and limiting frequency, which are integral to his system. Then, I will reconstruct von Mises' argument as to why single-case probabilities must be nonexistent. Finally, I will offer an objection to his line of reasoning which

exposes the inadequacies of his theory of probability, as well as a possible defense von Mises may make.

For the sake of mitigating convolution, von Mises carefully defines the terms used in his theory, as well as specifying the types of events his theory applies to. Von Mises introduces a concept that "denotes a sequence of uniform events or processes which differ by certain observable attributes, say colours, numbers, or anything else" which he calls "the collective" (von Mises, 11). The collective is also referred to as a reference class, and it lays the basis for von Mises theory. As stated, a collective may be a sequence of events, such as the eye colors of sibling children, as well as processes, like the set of all fair dice rolls. Within these collectives are the observable attributes, which are the results of events or outcomes of processes, such as blue eyes or a die landing on six. Von Mises stresses the necessity of a collective existing before probability can even be discussed. He states "the principle which underlies the whole of our treatment of the probability problem is that a collective must exist before we begin to speak of probability" (12). Therefore, von Mises' definition of probability discussed hereinafter is only concerned with "'the probability of encountering a certain attribute in a given collective'" (12). The methodology von Mises introduces requires what he calls "unlimited repetition;" that is, mass phenomena or repetitive events that can theoretically be repeated an infinite number of times. He uses this concept in conjunction with the "limiting frequency" in order to find the objective probability of an event. The limiting frequency is the value between 0 and 1 inclusive of an attribute being the case given an infinite collective. For instance, the limiting frequency of a coin flip landing heads is the value that is converged upon for a fair coin to land heads if flipped an infinite number of times. This value is found as a ratio of favored attributes to number of trials, which for heads on a coin flipped the first ten times of infinite trials, could be this

sequence: 1/1, 1/2, 1/3, 2/4, 3/5, 4/6, 4/7, 4/8, 4/9, 5/10... The intuition behind the unlimited repetition is that as the number of trials approaches infinity, the ratio stabilizes and converges to the objective probability. The limit arrived upon by the non-infinite sequence of ratios is referred to by von Mises as the relative frequency, and it is this relative frequency from which the limit is extrapolated. Von Mises further clarifies his definition of a collective by adding that collectives must display sufficient reasons to believe that the relative frequency of the observed attribute would tend to a fixed limit if the observations were indefinitely continued. This limit will be called the probability of the attribute considered within the given collective.

Von Mises asserts “It is utter nonsense to say, for instance, that Mr. X, now aged forty, has the probability 0.011 of dying in the course of the next year” (17-18). This is his dogmatic dismissal of the possibility of single-case probabilities. His argument against this possibility is as follows:

1. Probabilities are determined by a limiting frequency.
2. Limiting frequencies are extrapolated from relative frequencies.
3. Relative frequencies represent a collective.
4. Collectives must display sufficient reason to believe that they tend to a fixed limit if the observations were infinitely continued.
5. Single-cases cannot display sufficient reason to believe that they tend to a fixed limit if the observations were infinitely continued by virtue of being a single case.
6. Single-case probabilities are not derived from collectives.

C: Single-case probabilities are not probabilities.

Von Mises' reasoning is valid, but not sound. It is valid that if A requires B and B requires C and C requires D, and E is not a D, then by transitivity E cannot be an A. However, despite this validity, von Mises' argument against single-case probabilities makes a misstep in its premises that compromise its soundness.

The assumption in von Mises' supporting premise that single cases cannot display sufficient reason to be a true probability proves to be misguided by one of von Mises' own examples. He cites the "great gambling banks of Monte Carlo and elsewhere" (16) as proof that the copiously large amounts of data of frequencies for probabilities utilized by these casinos to remain profitable is sufficient enough belief in a limiting value. Had these casinos not held this belief, they surely would lose money. The millions of data points in the sequences of the relative frequencies of individual gambling games create a large enough collective for this sufficient reason. So, as my objection goes, because von Mises concedes this method of profitability from casinos is precipitated on sufficient evidence to trust the probabilities of their games, I raise an argument for the sufficiency of single-cases to also have the opportunity to have probabilities. My contention is based on the role of casino bookmakers. Take, for instance, a boxing match between two boxers who have never fought each other prior. The outcome of this event falls under the category of events that von Mises allows for probabilistic analysis due to the fact that casinos consistently profit off of such odds similarly to the casino games mentioned by von Mises, and furthermore insurance companies essentially are betting on the health of their customers, which von Mises deems adequate for analysis as well. This is unique enough of a scenario to be considered a single-case, yet despite von Mises' claim of the nonexistence of single-case probabilities, casino bookmakers must still draw up betting lines good enough for the casino to turn a profit. If the bookmakers have no collective sufficient enough to determine a

limiting frequency from, they must utilize information and data from other sources, such as the fighter's records, public favorites, training regimens, and countless other data points. It is important to emphasize that the bookmakers are not attempting to predict who will win a fight, but rather are drawing up odds that the casino can turn a profit on without prior knowledge of past odds for a single case. This does not deny, however, that these betting odds are dependent on various single-case probabilities, including who is predicted to win but also probabilities such as which fighter bettors are more likely to favor. Therefore, there must exist sufficient reasons other than solely the relative frequencies of collectives to draw probabilities for single cases from, or else no casino would offer betting odds on bouts between boxers who have never fought lest they risk losing money. Furthermore, the possibility that casinos may recoup any losses taken on by poor bookmaking through other avenues at a boxing event such as refreshments and ticket sales does not disprove the necessity of bookmakers to ascertain accurate odds due to the existence of gambling platforms through which gambling is their only source of revenue.

I will now address what I deem to be the strongest rebuttal to my objection and further discuss how such a response paradoxically undermines von Mises' own argument. One may claim that there exist no single-case probabilities not because they are nonsensical, but because every event that von Mises claims to be of the type for which a probability can be calculated exists as a value in some collective. In other words, the bout between two fighters who have never fought is not a unique single case, but rather could be part of various collectives such as 'fights between a boxer of record W_1-L_1 and a boxer of record W_2-L_2' or 'fights between boxers favored by the crowd 20/80' or even 'fights between boxers of height H_1 and weight W_1 and height H_2 and weight W_2' as well as numerous other collectives. Von Mises concedes that the restriction of collectives in an attempt to define a single-case probability will only result in the

single case itself, thus dismissing it as nonsense. However, this restrictive process seems to be akin to a bookmaker's process of establishing odds, as they have no other information to utilize. In fact, I would go so far to say that a probability can be established from the collective of bookmaker odds set on single-case sporting events by a limiting value to prove that bookmakers have a remarkably high probability of setting profitable odds, but I digress. Returning to my response, von Mises must concede that "Mr. X" can be sensibly assigned a probability of death either because he is a single case and my argument for single-case probabilities is convincing, or because just as there is extensive evidence for a coin having a limiting frequency of 1/2 due to the collective that flipping a coin is a part of, there must be extensive evidence of Mr. X's probability of death in the next year being 0.011 due to his membership in a collective, therefore not making Mr. X a single case. Whether or not there exist single-case probabilities, the probability of Mr. X's death cannot be nonsensical.